Improving CPU Efficiency for Data-Intensive Applications through Cache-Aware Spill Optimization

Register Spill Behaviour and Cache Hit/Miss Rates on RISC-V SPEC CPU2017

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gem5 TimingSimpleCPU · workload profile `test` · L1D 4 KiB, L1I 32 KiB, L2 32 KiB, direct-mapped L1D/L2 · gem5 `e0a12aa4e3+dirty`

Definitions (metric schema 9). Every stack store issued inside the region of interest, and every later load that reads it back (the load's address lies in the bytes the store wrote; widths may differ), belongs to exactly one class. `spill`: register-allocator spill code, identified by GCC itself (`-dP` RTL dumps of the same build, memory operands in the LRA spill area). `csave`: callee-saved register save/restore, return address included, identified from the binary's CFI (`.eh_frame`). `other_stack`: every remaining SP/FP-relative stack store and reload (stack-passed arguments, memory-resident locals). `stack_*` is the sum of the three. For each class an *event* is one (store, first reload) pair, so `event` ≤ `store` and `load` ≥ `event`. All counters cover only the region delimited by `m5_work_begin/m5_work_end`: the gem5 statistics are dumped and reset at those boundaries, so the cache columns span the same window, and the cache tags carry the class, so per-class hit/miss rates come straight from gem5. *Spill %* and *Csave %* are the class's store+reload instructions over `ROI_insts`; *spill/stack %* is the share of stack events that are real spills.

Benchmark	ROI_insts	stack_event	**spill_event**	csave_event	other_event	spill/stack %	Spill %	Csave %	L1D m %	L2 m %	sp_load m %	sp_store m %	cs_load m %	cs_store m %
cactuBSSN	48 023 254 960	3 820 802 773	3 797 752 332	22 956 427	94 014	99.4	21.32	0.10	17.357	48.80	6.416	5.506	1.503	0.285
parest	37 262 401 281	130 109 038	16 148 624	110 058 554	3 901 860	12.4	0.22	0.59	15.359	47.58	20.704	6.626	4.432	5.347
deepsjeng	46 202 686 169	1 453 401 396	380 604 124	1 046 384 737	26 412 535	26.2	2.78	4.53	13.686	19.74	6.735	9.452	4.189	4.775
leela	26 063 891 644	808 702 134	26 356 444	749 189 404	33 156 286	3.3	0.36	5.75	8.956	17.75	19.596	0.455	3.522	4.806
namd	31 914 751 099	336 702 115	334 454 030	1 939 340	308 745	99.3	3.13	0.01	15.582	37.33	12.114	2.494	4.501	4.822
lbm	6 506 800 040	274 131 044	274 130 500	544	0	100.0	9.67	0.00	27.221	75.06	1.630	5.456	15.074	0.184
nab	6 691 483 051	152 456 167	124 535 928	24 348 560	3 571 679	81.7	7.90	0.73	10.786	15.30	1.540	1.648	2.584	2.927
xz	2 014 351 521	34 435 119	4 683 669	26 392 608	3 358 842	13.6	0.68	2.62	8.704	28.28	1.326	0.574	2.159	2.774
xz_4m_l1	2 001 462 938	30 569 523	3 959 703	23 548 312	3 061 508	13.0	0.59	2.35	10.079	36.30	1.450	0.593	2.726	1.808
xz_4m_l2	2 083 103 943	29 428 092	3 700 891	22 776 791	2 950 410	12.6	0.54	2.19	11.652	42.38	1.490	0.611	3.599	2.134
xz_4m_l3e	10 573 516 663	134 644 921	61 575 948	66 069 008	6 999 965	45.7	2.51	1.25	9.401	25.69	3.079	2.596	3.576	3.921
xz_4m_l4	4 011 967 585	68 464 242	20 213 033	44 411 548	3 839 661	29.5	2.12	2.21	10.814	33.53	3.070	3.152	3.148	2.674
xz_4m_l4e	12 115 711 352	137 799 962	63 036 496	67 532 844	7 230 622	45.7	2.28	1.11	9.165	24.56	3.194	2.620	3.659	4.145
xz_1m_l0	711 271 503	13 362 560	1 091 347	10 938 543	1 332 670	8.2	0.62	3.08	7.980	29.63	2.021	0.928	1.658	1.510
xz_1m_l1	719 848 327	12 739 426	973 549	10 458 508	1 307 369	7.6	0.58	2.91	8.793	33.78	2.138	0.935	1.717	1.579
xz_1m_l2	724 690 222	12 498 311	956 442	10 241 945	1 299 924	7.7	0.57	2.83	8.922	34.53	2.151	0.939	1.797	1.667
xz_1m_l3e	1 994 406 562	34 952 797	12 876 553	20 010 020	2 066 224	36.8	2.59	2.01	8.680	23.89	2.686	3.225	1.979	3.759
xz_1m_l4	1 076 057 785	23 230 841	5 156 845	16 486 476	1 587 520	22.2	2.01	3.06	8.845	28.51	2.900	4.939	1.754	3.271
xz_1m_l4e	2 198 787 666	35 410 889	13 089 895	20 222 586	2 098 408	37.0	2.43	1.84	8.884	22.40	2.786	3.234	2.007	3.876
imagick	117 733 190	2 444 490	2 450	2 437 082	4 958	0.1	0.02	4.14	7.434	47.92	16.947	70.350	13.154	0.436
blender	994 421 777	46 417 461	4 430 041	40 844 530	1 142 890	9.5	1.16	8.21	9.917	31.78	2.962	7.024	5.637	4.108
povray	2 241 026 793	168 869 468	2 121 674	155 853 333	10 894 461	1.3	0.27	13.91	13.662	37.82	13.927	21.862	3.772	4.330
omnetpp	12 426 246 371	842 780 719	5 233 033	791 025 233	46 522 453	0.6	0.09	12.73	15.055	36.64	0.724	10.668	3.306	3.556
xalancbmk	395 919 938	18 468 307	295 564	17 282 226	890 517	1.6	0.23	8.73	15.161	41.74	13.188	18.142	4.760	6.007
x264	475 535 613 470	19 842 562 937	13 777 288 367	5 322 257 634	743 016 936	69.4	6.24	2.24	3.981	26.05	0.981	1.203	1.895	2.419
gcc	16 026 850	1 021 294	52 739	952 288	16 267	5.2	0.84	11.88	10.137	32.44	2.274	4.977	3.935	2.807
mcf	24 426 982 047	234 360 896	91 410 910	142 147 076	802 910	39.0	1.58	1.16	28.220	72.37	8.955	10.166	5.465	6.451
perlbench_makerand	2 421 667 618	224 582 067	55	224 562 979	19 033	0.0	0.08	18.55	12.684	0.38	0.000	1.444	14.407	5.926
bwaves_1	17 288 358 363	145 483 800	145 451 376	26 481	5 943	100.0	3.55	0.00	44.118	27.18	5.350	1.371	4.864	6.053
bwaves_2	17 287 430 556	145 483 720	145 451 376	26 421	5 923	100.0	3.55	0.00	44.120	27.18	5.350	1.371	4.879	6.060
exchange2	57 667 143 431	1 874 880 445	1 304 771 747	345 493 349	224 615 349	69.6	5.03	1.20	4.008	1.04	3.416	3.666	4.345	3.792
fotonik3d	3 098 360 920	28 397 309	27 853 226	437 206	106 877	98.1	3.84	0.03	22.434	38.96	9.885	0.425	2.099	4.269
roms	28 955 766 090	203 007 943	182 777 748	12 238 510	7 991 685	90.0	4.79	0.08	30.012	38.90	7.109	3.376	3.203	4.921
wrf	84 396 927 626	420 516 285	109 794 207	285 403 859	25 318 219	26.1	0.91	0.68	26.370	26.50	13.692	9.129	4.905	2.812
cam4	33 466 803 465	585 237 538	168 288 974	404 426 589	12 521 975	28.8	2.11	2.42	14.018	44.30	5.427	4.410	2.154	2.157
Total	1 003 626 872 816	32 328 356 029	**21 110 519 840**	10 039 381 551	1 178 454 638	65.3	5.17	2.00	0.000	—	—	—	—	—